

Math 242, S09
Quiz 5

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the following logistic equation:

$$\frac{dP}{dt} = 2P(5 - P), \quad P(0) = 2.$$

(a) Solve the above initial value problem for $P(t)$.

Solution: $\int \frac{1}{P(5-P)} dP = \int 2 dt$

$$\int \left(\frac{1}{5} \right) \left(\frac{1}{P} + \frac{1}{5-P} \right) dP = 2t + C$$
$$\int \left(\frac{1}{P} + \frac{1}{5-P} \right) dP = 10t + C$$
$$\ln|P| - \ln|5-P| = 10t + C$$
$$\left| \frac{P}{5-P} \right| = e^{10t+C}$$
$$\frac{P}{5-P} = Ae^{10t}$$
$$P(0)=2 \Rightarrow \frac{2}{5-2} = A \Rightarrow A = \frac{2}{3}$$

Thus, $\frac{P}{5-P} = \frac{2}{3} e^{10t}$

$$P(t) = \frac{10e^{10t}}{3+2e^{10t}}$$
$$= \frac{10}{2+3e^{-10t}}$$

(b) Find the limit $M = \lim_{t \rightarrow +\infty} P(t)$.

$$M = \lim_{t \rightarrow +\infty} \frac{10}{2+3e^{-10t}} = \frac{10}{2} = 5$$